

Liquid-Liquid Extraction

Final Lab Report
Unit Operations Lab 5
April 6, 2026

Group 2
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1. Executive Summary (10 pts)

In this experiment, liquid–liquid extraction was used to remove acetic acid from mineral oil into water using a three-stage mixer settler system. The goal was to study how agitator speed affects the separation and to calculate stage efficiencies.

Acetic acid concentrations in both phases were measured using titration with NaOH. The results showed that the concentration decreased from stage 1 to stage 3, which means the extraction was working as expected. This trend was seen for all agitator speeds. However, mixing speed had a noticeable effect, with higher rpm removing more acetic acid between stages. This indicates that increasing agitator speed improved contact between the two phases and helped improve mass transfer.

At 400 rpm, the raffinate concentration dropped from 1.051 wt% in stage 1 to 0.055 wt% in stage 3. When the speed was increased to 600 and 800 rpm, the final concentrations were lower, showing that better mixing improved the extraction performance.

McCabe–Thiele analysis gave about 1 theoretical stage, while the system had 3 actual stages. This corresponds to an overall efficiency of 33.3 percent. The Murphree efficiencies ranged from 7.09 percent to 39.58 percent depending on the stage and speed, which shows that equilibrium was not fully reached in each stage.

Overall, the results followed expected liquid–liquid extraction behavior, but the values were lower than ideal. This is likely due to mixing limitations, short contact time, and experimental errors.

2. Introduction (10 pts)

Liquid–liquid extraction is a process used to remove a substance from one liquid by transferring it into another immiscible liquid. In this experiment, acetic acid is removed from mineral oil using water as the solvent. Since water and oil form two separate phases, the acetic acid distributes between them until equilibrium is reached, resulting in a stable concentration ratio between the phases. The goal of the extraction process is to transfer as much acetic acid as possible into the aqueous phase. To improve separation, the process is carried out in multiple stages rather than a single step, where each stage consists of a mixer for mass transfer and a settler for phase separation [1].

An important factor in this process is the mixing intensity, which is controlled by the agitator speed. The agitator speed affects how well the two phases come into contact. If mixing is too slow, there is insufficient contact and mass transfer is limited. If mixing is too fast, the phases may not separate properly. Therefore, an optimal range of mixing exists that promotes mass transfer while maintaining clear phase separation. As a result, agitator speed has a direct impact on the efficiency of the extraction process. Liquid–liquid extraction is widely used in industry for the separation and purification of compounds, such as removing organic acids from hydrocarbon streams and treating wastewater. These applications rely on efficient mass transfer and proper mixing conditions to achieve effective separation [2].

The objective of this experiment is to use a three-stage mixer-settler system to extract acetic acid from oil using water as the solvent. By operating the system at different agitator speeds while keeping other variables constant, the effect of mixing on acetic acid transfer and stage efficiency can be evaluated. The results provide insight into how operating conditions influence the performance of staged liquid–liquid extraction systems [1].

3. Theory (10 pts)

Liquid-liquid extraction is a separation process in which a solute is transferred from one liquid phase to another, in which it is immiscible, based on differences in solubility. In this experiment, acetic acid is extracted from a mineral oil phase (raffinate) into water (extract). Because these two liquids are almost entirely immiscible, they form separate phases, allowing the solute material to distribute between them according to its equilibrium.

A multistage, countercurrent extraction system is used, as shown in the figure below:

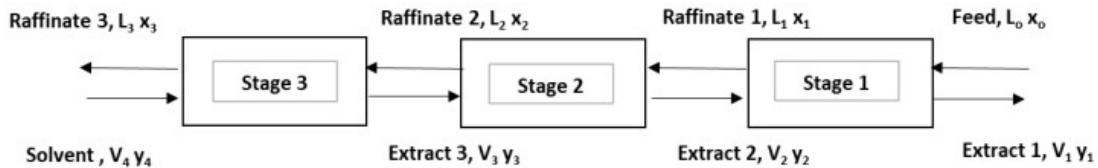


Figure 1 - Counter-Current 3-Stage Process Flow Diagram [1]

The raffinate and extract streams flow in opposite directions across three mixer-settler stages. This configuration maximizes the driving force for mass transfer in the system and improves the overall separation efficiency.

At equilibrium, the concentrations of acetic acid in each phase are related by the distribution relationship. For a linear system:

Where x is the solute concentration in the raffinate phase, y is the concentration in the extract phase, and K is the equilibrium distribution coefficient [1]. A larger K indicates that the solute preferentially transfers into the solvent phase, improving overall extraction.

In a single stage, a solute mass balance ensures that the amount removed from the raffinate equals the amount transferred to the extract. If one extends this to the multistage systems, a relationship for the solute concentration leaving hypothetical stage $n+1$ is equal to:

Where the extraction factor, E , is defined as:

In this equation, L is the raffinate flow rate, and V is the solvent flow rate. Increasing E , either through higher solvent flow or from a favorable equilibrium, thus leads to greater solute removal in the system.

Additionally, each stage consists of a mixer and a settler. In the mixer, the two phases are contacted using an agitator to increase interfacial area and enhance mass transfer. The mixture then enters the settler, where the phases separate based on density differences, with the lighter phase forming the top layer and the heavier phase settling at the bottom. The performance of each stage depends on how closely equilibrium is reached. Agitator speed directly affects this, as higher speeds generally improve mass transfer but may also influence phase disengagement.

The number of theoretical stages required for the desired separation can be determined graphically using the McCabe-Thiele approach. This involves plotting the equilibrium data/relationship between y_A and x_A along with an operating line and then taking “steps” between the two curves on the graph to count the number of stages. An example of the method can be seen below:

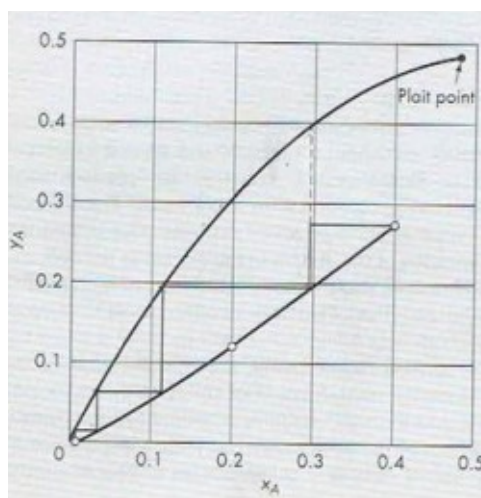


Figure 2 - McCabe-Thiele Diagram for Extraction [1]

This graphical method provides a direct way to compare experimental performance with the ideal stage behavior of the system and can be used in this particular experiment to determine the required number of stages needed for separation.

Now, looking at stage efficiency, one must understand that the real stages do not reach perfect equilibrium; thus, efficiencies are used to quantify separation performance. The Murphree efficiency for the extract phase is defined as:

And for the raffinate phase:

Where the starred values represent equilibrium compositions. Lastly, the overall stage efficiency is given by:

Where N is the actual number of stages and N_{th} is the number of theoretical stages.

Application to Experiment

In this experiment, the acetic acid concentrations in both phases are determined by titration with standardized 0.5 M NaOH. These measurements are used to:

- Construct equilibrium and operating diagrams
- Perform McCabe-Thiele graphical analysis to determine theoretical stages
- Calculate Murphree and overall efficiencies
- Analyze the effect of agitator speed on extraction performance

Agitation will play a critical role in achieving near-equilibrium conditions during acetic acid separation. Increasing impeller speed should, in theory, enhance mass transfer by increasing turbulence and increasing the interfacial contact area. However, excessively high or low speeds may reduce performance by either limiting mixing or hindering phase separation.

Table 1 – Variables & Units

Variable	Units
	Mass fraction or wt%
	Mass fraction or wt%
K	Dimensionless
L	kg/s or gpm
V	kg/s or gpm
E	Dimensionless
	Mass fraction (-)
	Mass fraction (-)
	Mass fraction (-)
	Mass fraction (-)
	Mass fraction (-)
	Mass fraction (-)
	Mass fraction (-)
	Dimensionless
	Dimensionless
	Dimensionless
	-
	-
n	-

4. Results (15 pts)

Acetic acid concentrations in the aqueous (extract) and organic (raffinate) phases were determined by titration with standardized NaOH. Since the reaction between acetic acid and NaOH is 1:1, the concentration of acetic acid in each sample was calculated from the titrant volume and then converted to weight percent based on the sample mass. The relation used was:

where V is the NaOH volume, M is the NaOH molarity, MW is the molecular weight of acetic acid, and m is the sample mass. For the aqueous phase, density was taken as 1 g/mL, while the oil density was taken as 0.87 g/mL.

This calculation is done in lab to determine how much acetic acid needed to be added to the feed tank, where the results showed an initial acetic acid concentration of 0.0254 wt. % and 764.2 mL of acetic acid needing to be added to the feed tank.

The calculated acetic acid concentrations in both phases for each agitator speed are summarized in Table 2 and plotted in Figures 3 and 4.

Table 1 – Acetic Acid Concentrations in Extract and Raffinate Phases

speed (rpm)	stage	X_A (oil wt%)	Y_A (water wt%)
400	1	1.051	0.1657
	2	0.120	0.0162
	3	0.055	0.0159
600	1	0.629	0.0729
	2	0.212	0.0138
	3	0.075	0.0115
800	1	1.201	0.0647
	2	0.150	0.0254
	3	0.080	0.0138

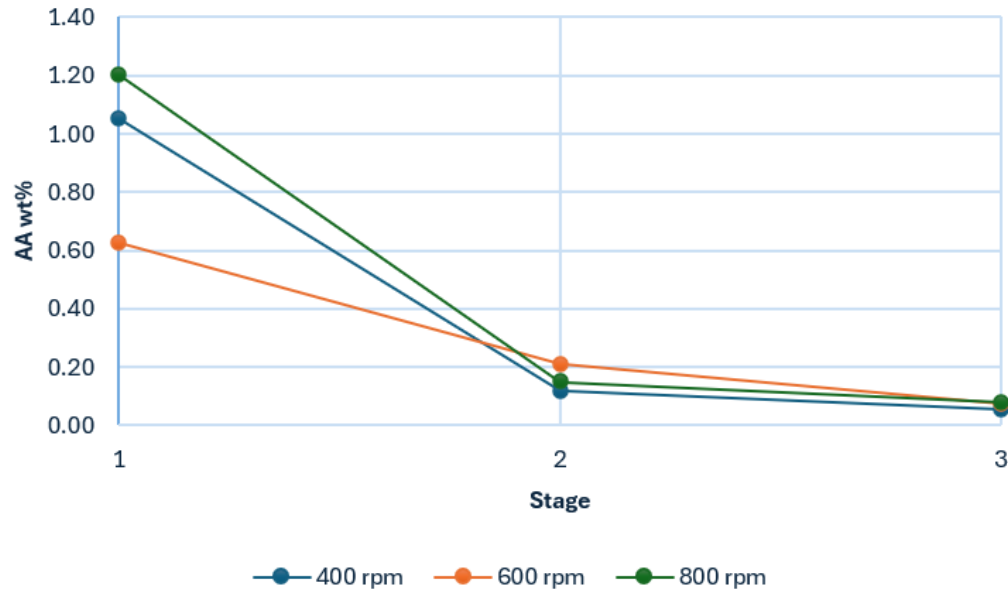


Figure 3: AA wt.% in water phase

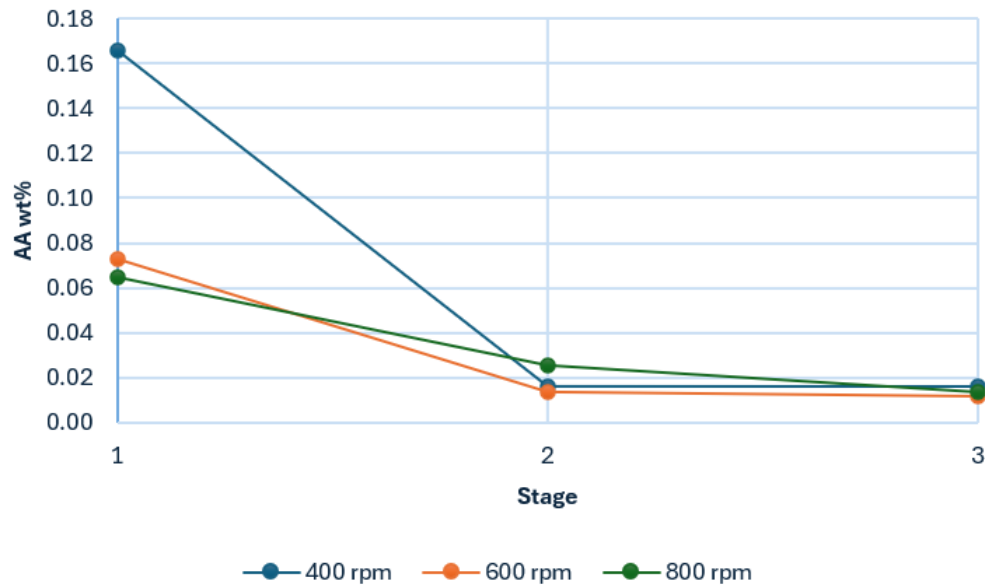


Figure 4: AA% in oil phase

In both phases, the concentration generally decreased from stage 1 to stage 3. Since stage 1 was closest to the feed side, this trend is consistent with progressive extraction of acetic acid through the stages. The general relationships appear reasonable, although some scatter is present, likely due to incomplete steady-state operation or titration endpoint uncertainty.

The acetic acid concentration in the extract phase was plotted against the concentration in the raffinate phase for each agitator speed, as shown in Figure 5.

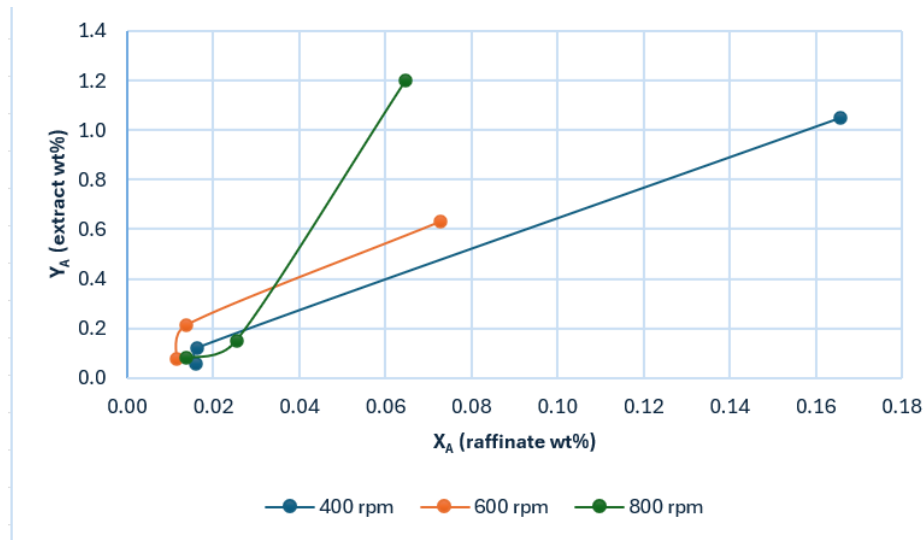


Figure 5: vs. for Different Agitator Speeds

This plot shows a positive relationship between X_A and Y_A , where higher acetic acid concentration in the oil phase corresponds to higher concentration in the water phase. This is consistent with the expected partitioning behavior of acetic acid between the two phases. The 400 rpm case shows the highest raffinate concentration, while the 600 and 800 rpm data are generally shifted toward lower raffinate concentrations, suggesting improved extraction at higher mixing speeds.

Experimental vs. data were compared with the equilibrium data from Appendix I, where a few of the higher points were removed to shorten the graph and make counting stages easier. A graphical McCabe–Thiele construction was performed for each agitator speed, and approximately one theoretical stage was obtained in each case, which can be seen in Figures 6-8 below:

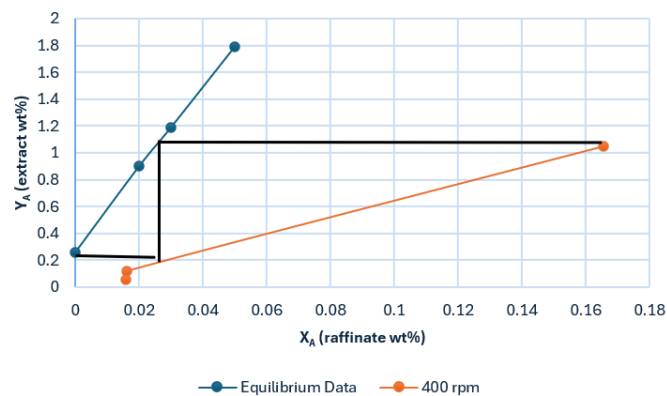


Figure 6: McCabe-Thiele diagram for 400 rpm

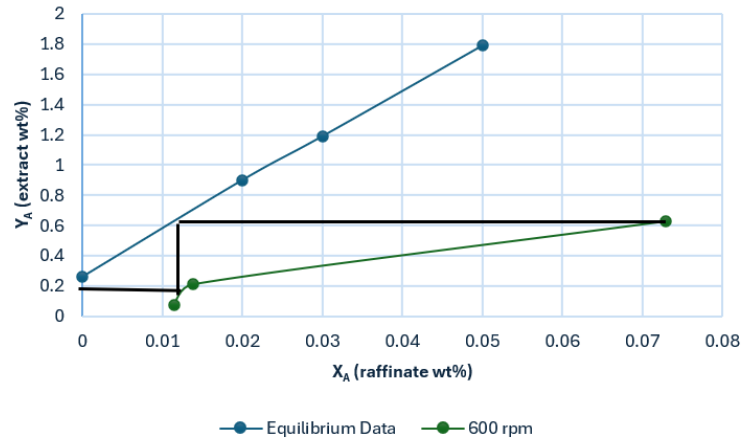


Figure 7: McCabe-Thiele diagram for 600 rpm

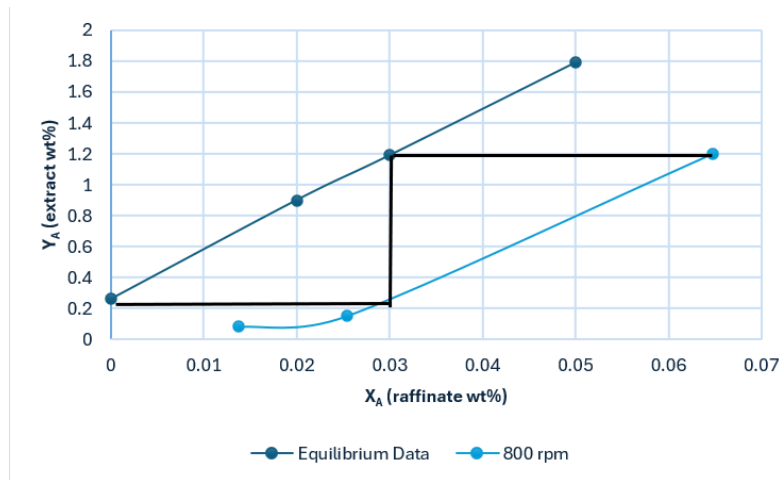


Figure 8: McCabe-Thiele diagram for 800 rpm

It is important to note that only the relevant – equilibrium data was used to avoid stretching the graph and to make the stage construction easier.

The extractor used three actual stages, so the overall efficiency was calculated as:

The experimental points followed the same general direction as the equilibrium relation, but were lower than the equilibrium curve. This suggests that the system followed the expected extraction trend, but did not approach equilibrium closely in each stage.

Extract-phase Murphree efficiencies were estimated using:

where was obtained from the equilibrium data corresponding to the measured raffinate composition. The calculated extract-phase Murphree efficiencies are shown in Table 3.

Table 3: Extract-phase Murphree efficiencies

Speed (rpm)	Stage 1	Stage 2	Stage 3
400	8.05%	9.03%	7.09%
600	12.59%	21.83%	11.95%
800	39.58%	7.17%	11.41%

The Murphree efficiencies were relatively low, which is consistent with the experimental data lying below the equilibrium curve. Raffinate-phase efficiencies were also attempted, but were found to be too inconsistent to report reliably. The overall efficiency from the McCabe–Thiele analysis was 33.3% for the three-stage extractor.

The results showed that acetic acid concentration generally decreased from stage 1 to stage 3 in both phases, indicating successful extraction across the mixer-settler system. The vs. plot followed the expected general relationship, though the experimental data were lower than the equilibrium data. McCabe–Thiele analysis gave approximately one theoretical stage, corresponding to an overall efficiency of 33.3%, and extract-phase Murphree efficiencies ranged from about 7% to 40% depending on stage and speed. Overall, the measured trends were physically reasonable, though the system appeared to operate somewhat below equilibrium conditions.

5. Discussion (15 pts)

The results from this experiment generally followed the expected behavior for a staged liquid-liquid extraction system. In both the aqueous and organic phases, the acetic acid concentration decreased from stage 1 to stage 3 at all agitator speeds, which indicates that the solute was progressively transferred from the oil phase into the water phase as the fluids moved through the extractor. For example, in the aqueous phase at 400 rpm, the concentration decreased from approximately 1.05 wt% in stage 1 to about 0.07 wt% in stage 3, showing a clear reduction in solute concentration across the system. This trend is consistent with the theory of countercurrent extraction, where each stage provides additional contact between the solvent and feed, allowing more solute to be removed from the raffinate stream [1], [3]. The direction of the concentration profiles therefore supports the idea that the system was operating as intended and that the stages were contributing to the overall separation.

The effect of agitator speed on extraction performance was also generally consistent with mass transfer theory. Higher agitator speeds produced lower acetic acid concentrations in the raffinate phase, especially in stage 1. For instance, the raffinate concentration at stage 1 was highest at 400 rpm (around 0.165 wt%), while the 600 rpm and 800 rpm cases showed lower values closer to 0.07 wt%, indicating improved extraction performance at higher mixing intensities. This behavior is expected because stronger agitation reduces the thickness of the mass transfer boundary layer and increases the interfacial area between the two liquids, allowing the solute to move more easily from one phase to the other [4]. Although the improvement was noticeable, the relationship between speed and efficiency was not perfectly linear, suggesting that other factors such as residence time or phase separation limitations may have influenced the results.

When the experimental data were compared with equilibrium behavior, the results showed reasonable agreement in terms of trends but not in magnitude. The experimental points followed the same general direction as the equilibrium curve, indicating that the system behaved according to expected partitioning relationships between the two phases. However, the data consistently fell below the equilibrium line, which suggests that the system did not reach equilibrium in each stage. The calculated overall efficiency of the system was approximately 33.3%, meaning that the three-stage extractor performed similarly to about one theoretical stage. This outcome is common in real extraction systems because perfect equilibrium requires sufficient mixing time, stable flow conditions, and efficient phase separation, all of which are difficult to achieve in a laboratory-scale unit [5].

The Murphree efficiencies calculated for individual stages were relatively low and varied between approximately 7% and 40%, depending on stage number and agitator speed. These values further support the conclusion that the system was operating below equilibrium conditions. Lower efficiencies are typically associated with incomplete mixing or insufficient residence time, both of which reduce the amount of solute transferred between phases during each stage. In practice, industrial extraction systems

often compensate for these limitations by increasing the number of stages or improving mixing conditions to achieve higher separation efficiency [3], [5].

Some aspects of the data also suggest possible issues with the experimental procedure. One likely factor was that the system may not have reached steady-state conditions before samples were collected. If steady state had not been fully established, the measured concentrations would not accurately represent the true operating conditions of the extractor. This would lead to variability in calculated efficiencies and could explain why some values appeared inconsistent between stages. Another possible source of error was the titration method used to determine acetic acid concentration. Identifying the exact endpoint of a titration can be difficult, especially when working with low concentrations or non-aqueous samples, and small differences in endpoint detection can produce noticeable changes in calculated results.

The assumptions used in the analysis were generally reasonable but still introduce some uncertainty into the results. For example, constant density values were assumed for both phases, even though the composition of each stream changed slightly from stage to stage. This simplification is commonly used in laboratory calculations and is unlikely to produce large errors, but it does contribute to some level of inaccuracy in the calculated concentrations. Another assumption was that the equilibrium data from the literature accurately represented the behavior of the system under experimental conditions. Since equilibrium relationships can vary with temperature and composition, any difference between the actual operating conditions and the reference data could affect the comparison between experimental and theoretical values [5].

Despite these limitations, several clear conclusions can be drawn from the experiment. The staged extraction system successfully removed acetic acid from the oil phase, as demonstrated by the consistent decrease in concentration across the stages. Increased agitator speed generally improved extraction performance by enhancing mass transfer between the phases. At the same time, the system did not reach equilibrium in each stage, which resulted in lower-than-ideal efficiency. These findings are consistent with established mass transfer and extraction theory and reflect the difference between theoretical predictions and real system behavior [1], [4].

Overall, the experimental trends were physically reasonable and aligned with the fundamental principles of liquid-liquid extraction. The decrease in concentration from stage to stage, the improvement in performance at higher mixing speeds, and the relatively low efficiency values all reflect the expected behavior of a laboratory-scale extraction system operating under non-ideal conditions. Future experiments could improve accuracy by allowing more time for steady-state conditions to be reached, repeating trials to confirm consistency, and using more precise analytical methods to reduce measurement uncertainty.

6. References (10 pts)

- [1] Department of Chemical Engineering. *Liquid-Liquid Extraction*. University of Illinois at Chicago, 2025.
- [2] C. J. King, *Separation Processes*, 2nd ed. McGraw-Hill, 1980.
- [3] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.
- [4] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.
- [5] P. C. Wankat, *Equilibrium-Staged Separations*. Prentice Hall, 1988.

7. Appendix I: Data Tabulation/Graphs (10 pts)

The raw measurements collected during the Liquid-Liquid Extraction experiments and further calculations on acetic acid concentrations and efficiencies are tabulated below:

Table 1 - Raw Experimental Data for Acetic Acid Calculations

Speed (rpm)	Stage	Water Volume (mL)	Oil Volume (mL)	NaOH (Water) (mL)	NaOH (Oil) (mL)
400	1	4	5	0.7	2.4
	2	5	8.5	0.1	0.4
	3	5.5	6.5	0.05	0.3
600	1	10.5	7.1	1.1	1.5
	2	8.5	10	0.3	0.4
	3	8	9	0.1	0.3
800	1	6	8	1.2	1.5
	2	8	9.5	0.2	0.7
	3	7.5	10	0.1	0.4

Table 2 – Equilibrium Data

XA (raffinate wt%)	YA (extract wt%)
0	0.26
0.02	0.9
0.03	1.19
0.05	1.79
0.06	2.39
0.07	3.27

Table 3 – Full Calculation Table

Speed (rpm)	Stage	X _A (oil wt%)	Y _{A,actual} (water wt%)	Y _{A,equilibrium}	E _{EM} %
400	1	0.166	1.051	11.69	8.05%
	2	0.016	0.120	0.78	9.03%
	3	0.016	0.055	0.77	7.09%
600	1	0.073	0.629	3.53	12.59%
	2	0.014	0.212	0.70	21.83%
	3	0.012	0.075	0.63	11.95%
800	1	0.065	1.201	2.81	39.58%
	2	0.025	0.150	1.06	7.17%
	3	0.014	0.080	0.70	11.41%

8. Appendix II: Error Analysis (10 pts)

Several sources of error caused uncertainty in the measured acetic acid concentrations and stage efficiencies. The main sources were titration error, sampling differences, and the system not running perfectly under ideal conditions.

The biggest source of error came from the titration. Since concentration depends directly on the amount of NaOH used, any small mistake in reading the burette will affect the result. If we assume a burette uncertainty of ± 0.05 mL, the percentage error depends on how much NaOH was used. For example, when the titration volume is very small, like 0.10 mL (which happens in later stages), even a small reading error can lead to a large percentage error.

Whereas for a larger volume such as 1.20 mL, the error decreases significantly:

This shows that samples with low concentration have much higher uncertainty, which helps explain why there is more scatter in the stage 2 and stage 3 data.

Another source of error comes from measuring the sample volume. If the sample volume is around 5 mL with an uncertainty of ± 0.2 mL, the percentage error is:

Since concentration depends on the sample volume, any error in measuring the volume will affect the result. This adds about $\pm 4\%$ uncertainty to the concentration values.

When we combine this with the other errors we can estimate the total uncertainty in the concentration as:

for low-volume titrations, and:

for larger titration volumes. This difference shows why the early-stage data is more reliable than the later-stage data.

Another source of error is that the system may not have fully reached steady state when samples were taken. Since steady state took about 10–17 minutes to reach, taking samples too early could cause around 10–15% error in the measured concentrations. This would make the measured values lower than expected and is one reason why the data falls below the equilibrium curve.

Another source of error is losing some acetic acid during sampling. For example, if about 0.001 g is lost from a sample that originally has around 0.006 g, the relative error is:

This loss is more important for low concentration samples and can make the measured values lower than they are.

Another source of error is cross contamination between stages. If a small amount of liquid, about 0.05 mL, is carried into a 5 mL sample, it causes about 1% contamination. Since concentrations between stages are different, this can lead to about a 2–5% error and make the separation look worse than it is.

Overall, the experiment showed that liquid–liquid extraction was successful in removing acetic acid from the oil phase. The concentration decreased across the stages, and higher agitator speeds generally improved the extraction performance, which agrees with expected mass transfer behavior.

Even though the system followed the theory closely, the results did not fully match equilibrium conditions. The experimental data fell below the equilibrium curve, and the overall efficiency was about 33%, with Murphree efficiencies ranging from about 7% to 40%. These values indicate that the system did not reach equilibrium in each stage.

The differences between experimental and theoretical results can be explained by sources of error such as titration, uncertainty, sampling errors, and the system not fully reaching steady state. Despite these limitations, the results were reasonable and consistent with the expected behavior of a multi-stage extraction system.

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9. Appendix III: Sample Calculations (10 pts)

Initial Acetic Acid Concentration in Oil Tank

Acetic Acid Concentration in Aqueous 400rpm Extraction

Acetic Acid Concentration in Oil 400rpm Extraction

Overall Efficiency of Separation

Murphree Extract Phase Efficiency

10. Appendix IV: Individual Team Contributions

	Time (Hours)			
	Joshua Olusoji	Omar Zaruk	Sanad Abu Hweij	Qussay Suleiman
Operator (Both Lab Days)	5	3	5	3
Pre-Lab Editing	1	1	1	2
Experimental Objective (Prelab)				1
Apparatus (prelab)	2			
Materials & Supplies (prelab)			1	
Procedure (prelab)		2		
Project Safety Evaluation (prelab)				2
Final Lab Editing	1	1		1
Executive Summary (final lab)			3	
Introduction (Final lab)				1
Theory (Final Lab)	2			
Results (final lab)				4
Discussion (final lab)		4		
References (final and/or prelab)	1	0.5		0.5
Data Tabulation/Graphs (final)	1			
Error Analysis (final lab)			4	
Sample Calculations (final Lab)	2			
Power Point Presentation	1	1	1.5	2
Total	16	12.5	15.5	17.5